

CASE STUDY / PANOLI UNIT • HPE SERVER • STORAGE • WINDOWS SERVER

Plant IT. A structured server build. A clear operating baseline.

PI Industries Limited | Panoli

HPE ProLiant DL360 Gen10 Plus

A detailed installation design covering rack mounting, redundant power, storage configuration, iLO management, Windows Server deployment and technical handover.

PO 4500606112 | 18 March 2025

8 cores

INTEL XEON 4309Y

32 GB

ORDERED RDIMM MEMORY

3 × 960 GB

ORDERED SATA SSDs

2 × 800W

HOT-PLUG POWER SUPPLIES

Project basis | Panoli requirement and server configuration

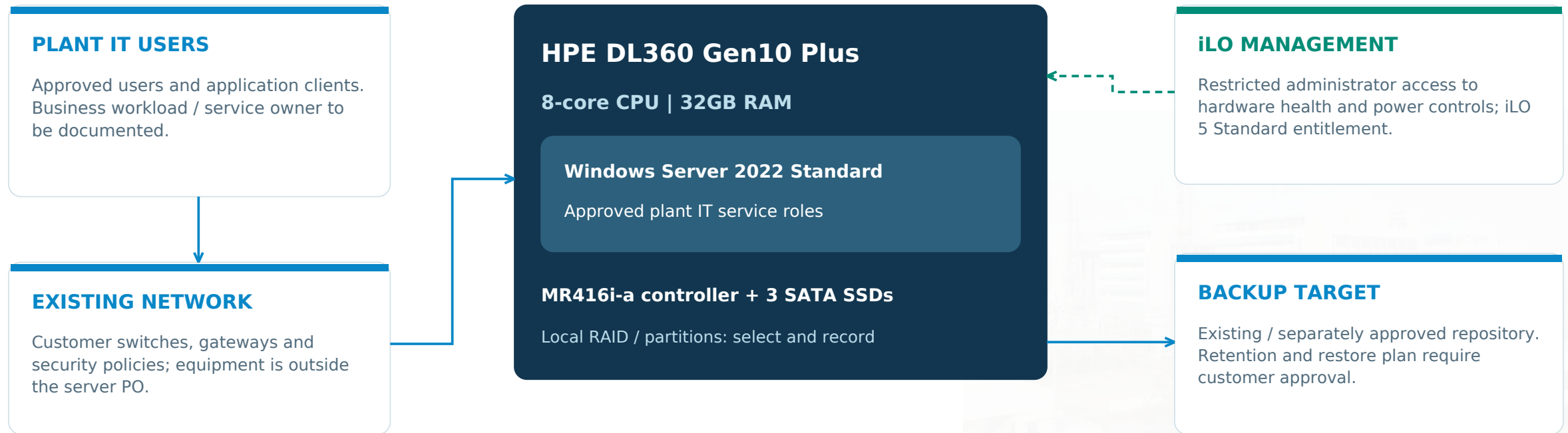
PI Industries’ Panoli order defines a single HPE server and Windows Server package. The case study turns that hardware scope into a step-by-step implementation design for approved plant IT services. Workload roles, installed RAID, network addresses and acceptance results are not established by the supplied order.

Component	Ordered specification	Build implication
PLATFORM / CPU	1 × DL360 Gen10 Plus NC; 1 × Xeon 4309Y, 2.8GHz, 8-core, 105W	Single physical server; verify exact product / serial IDs and firmware support.
MEMORY	1 × 32GB RDIMM 2R 3200	One populated DIMM. Record actual operating speed and slot placement; 3200 is the module rating.
STORAGE	8SFF Tri-mode U.3 backplane; MR416i-a 4GB + Smart Storage Battery; 3 × 960GB SATA MU SED SSD	2.88TB raw decimal capacity. RAID, logical drives, encryption and usable volume layout require approval.
CONNECTIVITY	BCM5719 four-port 1Gb BASE-T OCP adapter; iLO 5 Standard	Four data NIC ports plus management design; no 10G adapter is ordered.
POWER / SOFTWARE	2 × 800W hot-plug PSUs; MS WS22 16C Std ROK	Windows Server 2022 Standard 16-core ROK line; map power feeds and verify OS activation / access licensing.
SUPPORT	3-year NBD support	Verify serial entitlement and start date. NBD support does not establish a guaranteed repair time.

03 Architecture | 04 Rack / firmware / iLO | 05 Storage | 06 Windows / network | 07 Commissioning | 08 Handover & support

Site: Plot 640, GIDC, Panoli, Dist. Bharuch, Gujarat 394116. The PO specifies delivery for PN06; no completed installation report was supplied.

Architecture | How the Panoli server connects and operates



BLUE: application / backup data | DASHED GREEN: hardware management. Backup target, core switch and application systems are not supplied in this PO.

One host is one compute failure domain. Dual PSUs and RAID improve component resilience; they do not create a server cluster or disaster-recovery site.

Installation | Rack, power, firmware and iLO baseline

Stage	Engineer execution	Record / acceptance gate
01 RECEIVE	Check chassis, CPU, DIMM, SSD, controller, NIC and PSU part / serial labels against the PO. Inspect transport condition.	BOM / serial register; condition photographs; discrepancy resolution.
02 RACK	Verify compatible rails, rack depth and clearances. Install the 1U chassis; retain airflow and label rack position.	Rack-U allocation, rail fastening and airflow checks.
03 POWER	Connect both PSU cords to approved feeds. Verify PSU health and redundancy; inspect rack earthing with site personnel.	PDU / circuit map. Separate feeds require actual upstream independence.
04 FIRMWARE	Check supported BIOS / iLO / controller / NIC versions for the selected OS. Apply approved firmware in a controlled window.	Version baseline, update logs and hardware health report.
05 iLO	Set approved management IP / DNS / time. Replace default access with named authorised accounts; restrict access to admin paths.	iLO access / alarm checks; secure credential handover.
06 BOOT	Select supported UEFI / boot settings and approved security controls. Verify the storage battery and controller before OS installation.	Boot / security policy and controller health captured.

iLO Standard is ordered. Any feature requiring iLO Advanced, remote-console entitlement or separate management software must be verified separately. The server is installed in the approved IT room / rack. No hazardous-area certification or production-control integration is established by this order.

PRE-INSTALL DECISION / RAW CAPACITY ≠ USABLE CAPACITY / BACKUP REQUIRED

Storage design | Three SSDs, RAID and volume layout

The three ordered 960GB SSDs provide 2.88TB raw decimal capacity. Select a RAID policy using workload write profile, required capacity and recovery expectations. The examples are alternatives for approval; neither is claimed as the installed configuration.

SSD 1
 960GB | SATA

SSD 2
 960GB | SATA

SSD 3
 960GB | SATA

MR416i-a + STORAGE BATTERY
 Validate controller / battery health and approved cache policy. Never enable an unsafe write-cache policy to obtain a benchmark result.

Option	Approx. usable capacity	Resilience / design implication
RAID 1 + hot spare	0.96TB before filesystem overhead	Two-drive mirror plus one spare. Capacity is lower; spare availability supports a rebuild after a failed member.
RAID 5 across 3 SSDs	1.92TB before filesystem overhead	One-drive fault tolerance. Rebuild exposure / write behaviour must be acceptable; no dedicated spare remains.
OS / DATA VOLUMES	Size from workload demand	Create approved logical drives / partitions; reserve growth and log space. Partitioning alone does not isolate physical failure.
SED / BACKUP	Policy-driven	SED capability does not prove encryption is enabled. Record keys / recovery method if enabled; keep separate, tested backups.

RAID creation / changes can destroy data: obtain the approved layout before initialisation. Actual Windows-reported capacity is lower due to units and overhead.

OS and network | Windows deployment and port mapping

Install the ordered Windows Server edition using supported media and controller / NIC drivers. Keep the initial build isolated until hostname, addressing, DNS, patching and access controls are validated. No VMware licence or multi-host virtualisation scope is present in this order.

Example connection	Purpose	Configuration / test
NIC 1 production path	Approved application / server VLAN	Site-assigned static IP, prefix, gateway and DNS; prove approved client service access.
NIC 2 redundancy option	Alternate uplink, if designed	Use supported OS / switch teaming only after validating both ends; four ports do not imply a configured team.
NIC 3 backup option	Dedicated backup path, if approved	Separate transport / policy where required; avoid unintended routing and verify backup connectivity.
NIC 4 reserved option	Spare / future approved use	Keep unneeded interfaces disabled or unconfigured; record any later assignment through change control.
iLO management	Restricted hardware access	Dedicated approved management path; no routine end-user exposure.

WINDOWS BUILD

Set hostname, time / DNS, approved roles, firewall and named admin access. Patch in a controlled window and capture activation / driver status.

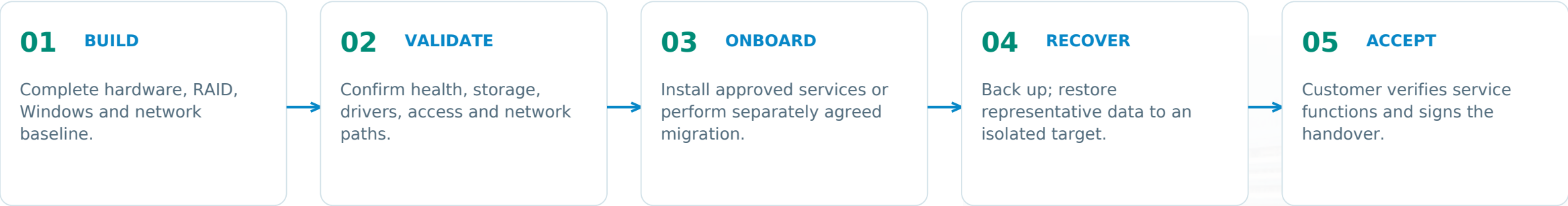
LICENSE CHECK

The 16-core ROK line covers the ordered 8-core server minimum on paper; verify ROK terms / activation and required CALs. No CAL quantity is stated.

NIC assignments are illustrative, not actual port labels or IP values. Confirm the final physical port-to-switch map and single default-route design.

Commissioning | Service onboarding, testing and acceptance

Bring the server into production only after the infrastructure baseline and customer service tests pass. If an existing workload is to be transferred, agree its supported migration method, backup and rollback path before cutover; migration is not a separate line item in the supplied PO.



Acceptance area	Test record / sign-off
HARDWARE / STORAGE	Healthy CPU / memory / drives / controller / battery; correct logical capacity; both PSUs recognised. Test power-feed loss only in an approved window.
OS / SERVICE	Windows activation, drivers, patch baseline, DNS / time, admin access and representative user application checks.
RECOVERY / CLOSEOUT	Backup success plus restore result; configuration record and customer-owned access. Record performance against agreed targets, not invented uptime figures.

If a cutover fails, isolate the new service, reconcile changes and restore the approved source / backup. Avoid duplicate IPs or two active copies of the service.

Operations and support | Panoli technical handover

Operating control	Recommended activity / evidence
DAILY HEALTH	Check iLO health / events, Windows services, volume free space and backup results; escalate repeated warnings.
PLANNED MAINTENANCE	Review capacity and approved firmware / OS patches; schedule changes with the service owner and preserve rollback records.
RECOVERY / ENTITLEMENT	Test restore per customer schedule; verify serial-based HPE support, Windows access licensing and backup retention.
HANDOVER PACK	PO / BOM, serials, rack / power / port maps, firmware versions, RAID / partition record, Windows build, backup / restore results and signed tests.

CUSTOMER & PROJECT

PI Industries Limited | Panoli unit
Plot 640, GIDC, Panoli, Dist. Bharuch, Gujarat 394116.
PO 4500606112 dated 18 March 2025.

D-TECH | CUSTOMER SUPPORT

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[HPE DL360 Gen10 Plus QuickSpecs](#)

[HPE MR controller guide](#)

[Microsoft server licensing](#)

Technical case study based on the server PO and supplied plant photo. The drawings explain an installation design; completion and test outcomes require project records.